

- 1 (a) (i) rate of change of angle / angular displacement swept out by radius M1
A1 [2]
- (ii) $\omega \times T = 2\pi$ B1 [1]
- (b) centripetal force is provided by the gravitational force B1
either $mr(2\pi/T)^2 = GMm/r^2$ or $mr\omega^2 = GMm/r^2$ M1
 $r^3 \times 4\pi^2 = GM \times T^2$ A1
 $GM/4\pi^2$ is a constant (c) A1
 $T^2 = cr^3$ A0 [4]
- (c) (i) either $T^2 = (45/1.08)^3 \times 0.615^2$ or $T^2 = 0.30 \times 45^3$ C1
 $T = 165$ years A1 [2]
- (ii) speed = $(2\pi \times 1.08 \times 10^8) / (0.615 \times 365 \times 24 \times 3600)$ C1
= 35 km s^{-1} A1 [2]
- 2 (a) atoms / molecules / particles behave as elastic (identical) spheres (1)